



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

03-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY
Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. A ball of radius R carries a positive charge whose volume charge density depends only on separation r from the ball's centre as $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$ where ρ_0 is constant. The electric field at distance r_1 from centre (For $r_1 < R$) is :-

(A) $\frac{\rho_0 r_1}{3\epsilon_0} \left(1 - \frac{3r_1}{4R}\right)$

(B) $\frac{\rho_0 r_1}{3\epsilon_0} \left(1 - \frac{r_1}{R}\right)$

(C) $\frac{\rho_0 r_1}{2\epsilon_0} \left(1 - \frac{r_1}{2R}\right)$

(D) None of these

2. A particle of mass m and carrying charge $-q_1$ is moving around a fixed charge $+q_2$ along a circular path of radius r . Find period of revolution of the charge $-q_1$ about $+q_2$.

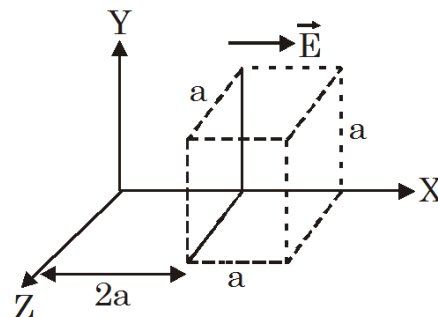
(A) $\sqrt{\frac{16\pi^3 \epsilon_0 m r^3}{q_1 q_2}}$

(B) $\sqrt{\frac{8\pi^3 \epsilon_0 m r^3}{q_1 q_2}}$

(C) $\sqrt{\frac{q_1 q_2}{16\pi^3 \epsilon_0 m r^3}}$

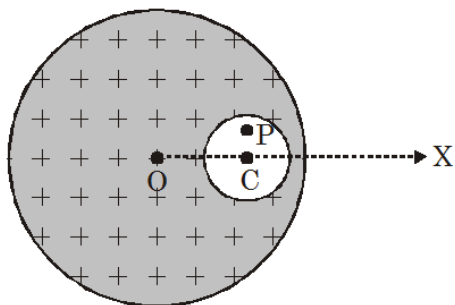
(D) Zero

3. In a region, electric field depends on X-axis as $E = E_0 x^2$. There is a cube of edge a as shown. Then find the charge enclosed in that cube.

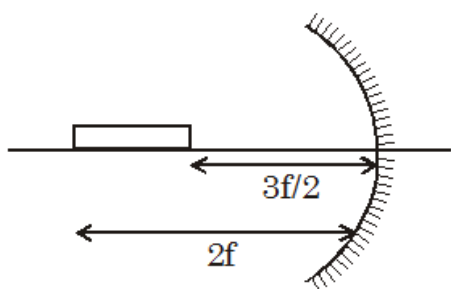


- (A) $5\epsilon_0 a^4 E_0$
- (B) $3\epsilon_0 a^4 E_0$
- (C) $4\epsilon_0 a^4 E_0$
- (D) Zero
4. A particle is performing SHM and its equation is $x = 10 \sin\left(4t - \frac{\pi}{3}\right)$, where time (t) is in second and x is the position from the mean position (in metre). The speed of particle at the instant when its displacement from mean position is 6m , is:-
- (A) $\frac{16\pi}{3} \text{m/s}$
- (B) $\frac{10\pi}{3} \text{m/s}$
- (C) 18m/s
- (D) 32m/s

5. A spherical cavity is made in a uniformly charged non-conducting sphere and a positive charge is released from position P as shown in figure. Then path followed by particle is :-

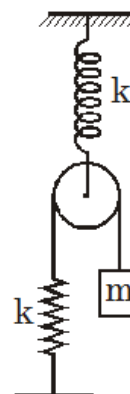


- (A) A straight line along radially inward of cavity
(B) A straight line along radially outward of cavity
(C) A straight line along positive X-direction
(D) A straight line along radially outward of sphere
6. A linear object is placed along the axis of a mirror as shown in the figure. If f is the focal length of the mirror then the length of image is

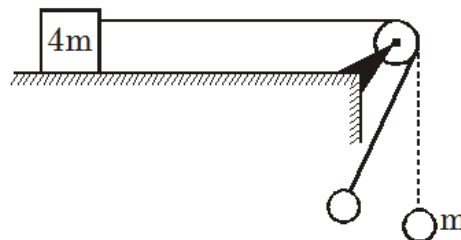


- (A) $\frac{2f}{3}$ (B) f (C) $\frac{f}{3}$ (D) $\frac{4f}{3}$
7. A pump of 200 W is lifting 2kg water per second from an average depth of 10m. The velocity with which water comes out of pump is :- (Take : $g = 9.8 \text{ m/s}^2$)
- (A) 2 m/s (B) 2.5 m/s
(C) 4 m/s (D) 3 m/s

8. A system is released from rest as shown in figure. Kinetic energy of mass m when it moves distance x is downward direction is (initially both spring are unstretched and all spring are massless)

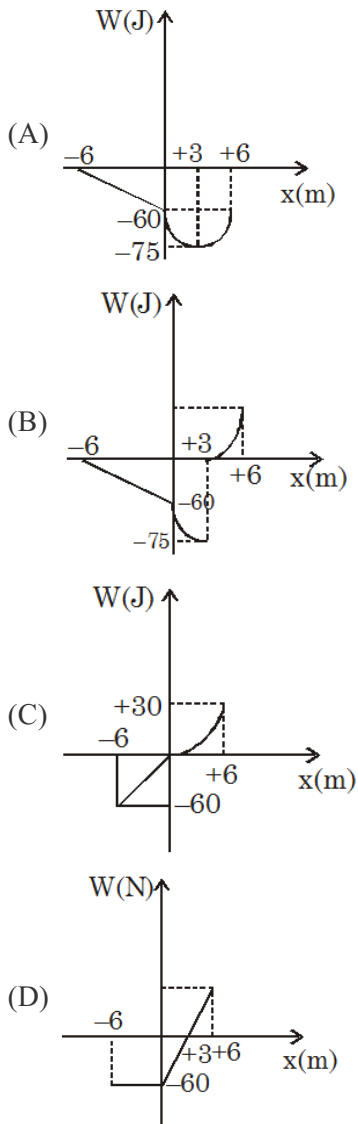
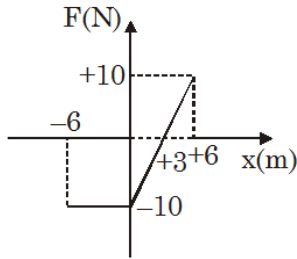


- (A) 10 mgx (B) $\frac{10\text{mgx} - kx^2}{10}$
(C) $\text{mgx} - kx^2$ (D) $\text{mgx} - \frac{kx^2}{2}$
9. Two bodies of mass m and $4m$ are attached by a string shown in the figure. The body of mass m hanging from a string of length ℓ is executing simple harmonic motion with small amplitude A while other body is at rest on the surface. The minimum coefficient of friction between the mass $4m$ and the horizontal surface must be :

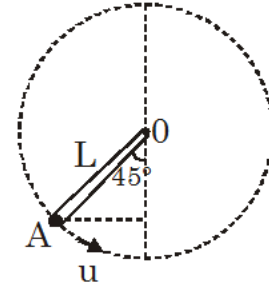


- (A) $\frac{1}{4} \left(1 - \frac{A^2}{\ell^2} \right)$ (B) $\frac{1}{4} \left(1 + \frac{A^2}{\ell^2} \right)$
(C) $\frac{1}{4} \frac{A}{\ell} \cos \theta$ (D) $\frac{1}{4}$

10. A particle is displaced from $x = -6\text{m}$ to $x = +6\text{m}$. A force F acting on the particle during its motion is shown in figure. Graph between work done by this force (W) and displacement (x) should be :-



11. A bob of mass m is attached to the lower end of a rigid light rod of length L and other end is fixed to a point O so that the system may rotate in vertical circle about O as shown in the figure. Initially the bob is at point A and rod makes an angle 45° from vertical. What minimum speed (v) should be imparted to the bob so that it may complete the circular track?



- (A) $u = \sqrt{gL(2 - \sqrt{2})}$
 (B) $u = \sqrt{gL(2 + \sqrt{2})}$
 (C) $\sqrt{gL(3 + \sqrt{2})}$
 (D) $\sqrt{gL(3 - \sqrt{2})}$
12. A force acting on a particle moving in the xy -plane is given by $F = (2y\hat{i} + x^2\hat{j})$ N, where x and y are in m. The particle moves along a straight line from the origin to $(5, 5)$. The work done by F is :-
- (A) 125 J (B) 66.7 J
 (C) 35 J (D) 25 J
13. If an astronomical telescope has objective and eye-piece of focal lengths 100 cm and 4 cm respectively, then the magnifying power of the telescope for the normal vision is :
- (A) 42 (B) 25 (C) 58 (D) 104

14. A particle in a certain conservative force field has a potential energy given by $U = \frac{20yz}{x}$.

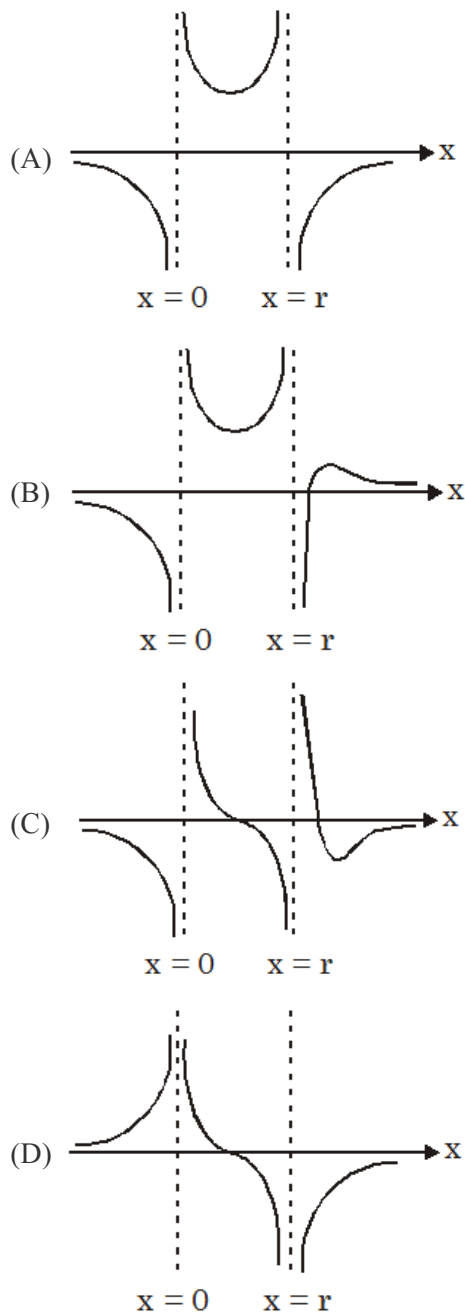
The force exerted on it is :-

- (A) $\left(\frac{20yz}{x^2}\right)\hat{i} + \left(\frac{20z}{x}\right)\hat{j} + \left(\frac{20y}{x}\right)\hat{k}$
 (B) $\left(\frac{20yz}{x^2}\right)\hat{i} - \left(\frac{20z}{x}\right)\hat{j} - \left(\frac{20y}{x}\right)\hat{k}$
 (C) $-\left(\frac{20yz}{x^2}\right)\hat{i} - \left(\frac{20z}{x}\right)\hat{j} - \left(\frac{20y}{x}\right)\hat{k}$
 (D) $-\left(\frac{20yz}{x^2}\right)\hat{i} + \left(\frac{20z}{x}\right)\hat{j} + \left(\frac{20y}{x}\right)\hat{k}$

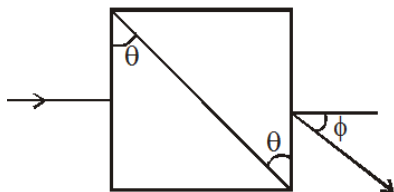
15. A convex lens A of focal length 20 cm and a concave lens B of focal length 5 cm are kept along the same axis with the distance d between them. If a parallel beam of light falling on A leaves B as a parallel beam, then distance d in cm will be :-

- (A) 25
 (B) 15
 (C) 30
 (D) 50

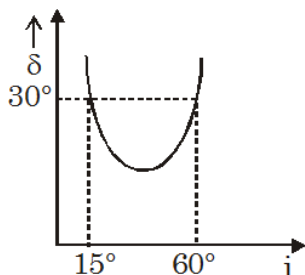
16. A positive point charge 'Q' is located at origin and a negative point charge 'q' is located on x-axis at a distance 'r' from 'Q'. Magnitude of charge of 'Q' is greater than that of 'q'. Then, which of the following may represent the E-x graph?



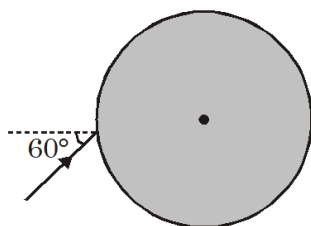
17. Two identical prisms with slightly different indices are located as shown. Angle θ is small. When a laser beam strikes one of the prisms perpendicular to the surface, the refracted ray is deviated by a small angle ϕ . The difference between the indices of refraction of the prism in terms of θ and ϕ is :-



- (A) $\frac{\phi}{\theta}$ (B) $\frac{\theta}{\phi}$ (C) $\frac{2\theta}{\phi}$ (D) $\frac{2\phi}{\theta}$
18. Figure shows a graph of angle of deviation δ versus angle of incidence i for a light ray striking a prism. Angle of prism is



- (A) 30° (B) 45° (C) 60° (D) 75°
19. A ray of light is incident on a glass sphere of refractive index $\sqrt{3}$ at an angle of incidence 60° as shown in figure. The total deviation suffered by the ray is



- (A) 30° (B) 60° (C) 90° (D) 120°

20. In an astronomical telescope, objective is of focal length 20 cm. What should be tube length for magnifying power of 10, under normal magnification ?

- (A) 10 cm (B) 22 cm
(C) 24 cm (D) 30 cm

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

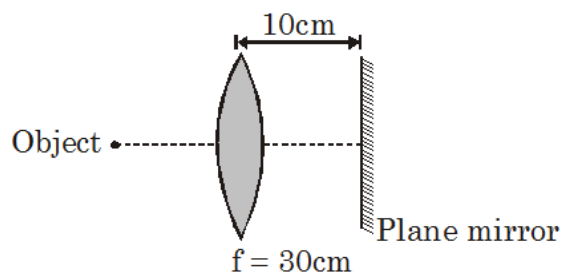
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

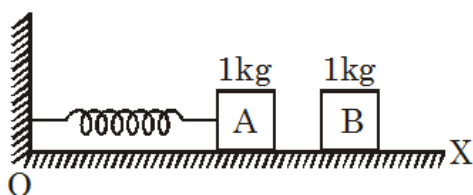
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

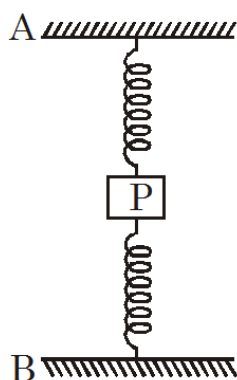
1. A sphere of radius R have volume charge density give as $\rho(r) = Kr$ for $r \leq R$ and 0 for $r > R$. If electric field at distance $\frac{R}{2}$ from centre is $\frac{KR^2}{N\epsilon_0}$, then N will be :
2. A point object is placed at distance of 20 cm from a lens as shown in figure. The distance of final image from the lens (in cm) is _____.



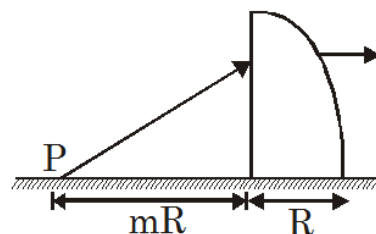
3. One end of an ideal spring is fixed to a wall at origin O and axis of the spring is parallel to x-axis. A block of mass 1 kg is attached to the free end of the spring and is performing SHM. Equation of position of the block in co-ordinate system shown in figure is $x = 10 + 3 \sin(10t)$, where t is in second and x in cm. Another identical block B, moving towards the origin with velocity 0.6 m/s collides elastically with the block A at $t = 0$. The new amplitude (in cm) of oscillations will be :



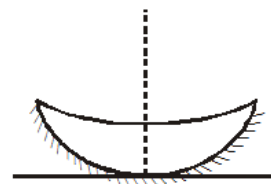
4. Two identical springs are attached to a small block P. The other ends of the springs are fixed at A and B such that AB is vertical. When P is in equilibrium the extension of top spring is 20 cm and extension of bottom spring is 10 cm. If the period of small vertical oscillations of P about its equilibrium position is $\frac{\pi}{N\sqrt{2}}$ sec, then fill the value of N.



5. A quarter cylinder of glass of radius R and refractive index 1.5 is placed on a table. A point object P is kept at a distance of mR from it. Find the value of $3m$ for which a ray from P will emerge parallel to the table as shown in the figure.

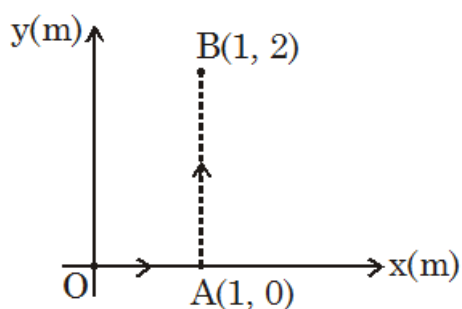


6. The convex surface of a thin concavo-convex lens of glass of refractive index 1.5 has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface. At what distance (in cm) should a pin be placed on optical axis so that image is formed at same place.

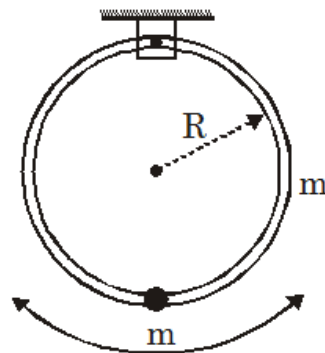


7. Two particles are performing simple harmonic motion with same amplitude and same frequency. When they are at same distance from mean position on opposite sides, their speeds are one fourth of maximum speed. Also at these positions their direction of velocities are opposite. If the phase difference between the two simple harmonic motions is $\frac{3\pi}{x}$, then find the value of x .

8. The distance between the object and its real image from the convex lens is 60 cm and the height of image is two times the height of object. The focal length of the lens (in cm) is f . Then $3f$ will be :
9. A variable force $\vec{F} = (x^2\hat{i} + xy\hat{j})$ N is acting on a particle which moves from point $(0, 0)$ to $(1\text{m}, 2\text{m})$ along the path OAB as shown. The work done by force is $\frac{p}{q}$ joules, where p and q are co-prime integers. Find the value of $(p + q)$.



10. A ring of mass m is pivoted at a point on its periphery and free to oscillates in vertical plane. A particle of same mass is attached at the lowest point of the ring then time period (in second) of small oscillations is [use $R = 2\text{m}$; $g = \pi^2 \text{ m/s}^2$]



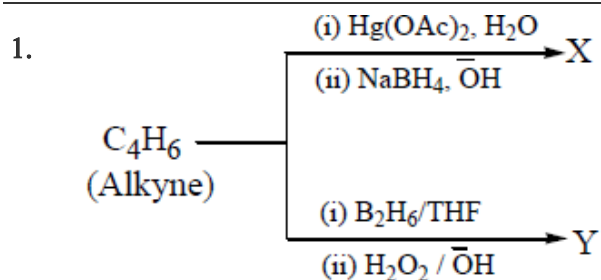
SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

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Zero Marks : 0 If none of the option is selected.

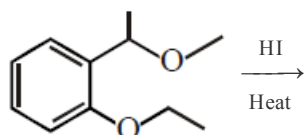
Negative Marks : -1 If wrong option is selected.



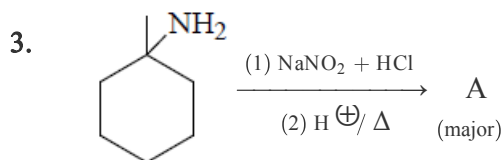
If X and Y are functional group isomers, then what is the structure of reactant (Alkyne).

- (A) $CH_3-C\equiv C-CH_3$
 (B) $CH_3-CH_2-C\equiv C-H$
 (C) $CH_2=CH-CH=CH_2$
 (D) $CH_3-CH=C=CH_2$

2. The major product formed in the following reaction is :

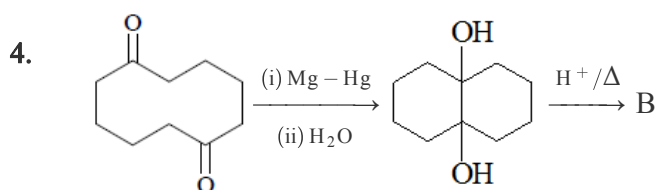


- (A) (B)
 (C) (D)



major product A is

- (A) (B)
 (C) (D) $CH_2=CH_2$



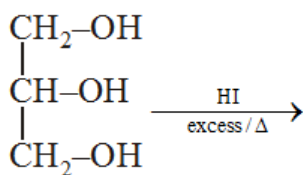
Final product (B) of the reaction sequence is :

- (A) (B)
 (C) (D)

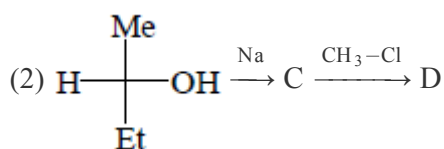
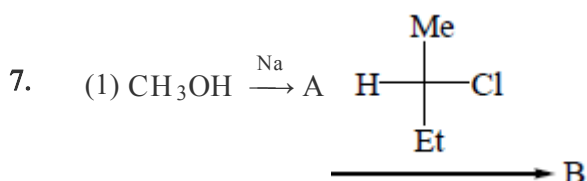
5. Which of the following alkene gives same product on reaction with HBr/CCl_4 and $HBr/H_2O_2/h\nu$

- (A) (B)
 (C) (D)

6. How many moles of HI will be consumed per mole of glycerol in the following reaction.



- (A) 4
(B) 5
(C) 6
(D) 7



Product B and D are :

- (A) Diastereomers
(B) Enantiomers
(C) Identical
(D) Functional group isomers

8. Match items of **List-I** with those of **List-II** :

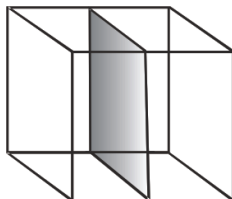
	List-I (Property)		List-II (Example)
(a)	Diamagnetism	(i)	MnO
(b)	Ferrimagnetism	(ii)	O ₂
(c)	Paramagnetism	(iii)	NaCl
(d)	Antiferromagnetism	(iv)	Fe ₃ O ₄

Choose the **most appropriate** answer from the options given below :

- (A) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)
(B) (a)–(iii), (b)–(i), (c)–(ii), (d)–(iv)
(C) (a)–(iii), (b)–(iv), (c)–(ii), (d)–(i)
(D) (a)–(iv), (b)–(ii), (c)–(i), (d)–(iii)
9. Two liquids A and B have, $P_A^0 : P_B^0 = 1 : 3$ at a certain temperature. Assume A and B form an ideal solution, and the ratio of moles of A to B in the liquid phase are 1 : 3, then mole fraction of A in vapour phase in equilibrium with the solution will be -
- (A) 0.1 (B) 0.2 (C) 0.5 (D) 1.0
10. The hydrolysis of an ester was carried out separately with 0.05 M HCl and 0.05 M H₂SO₄. Which of the following will be true ? [k = first order rate constant of reaction]
- (A) $k_{\text{HCl}} > k_{\text{H}_2\text{SO}_4}$
(B) $k_{\text{HCl}} < k_{\text{H}_2\text{SO}_4}$
(C) $k_{\text{HCl}} = k_{\text{H}_2\text{SO}_4}$
(D) $k_{\text{HCl}} = 2k_{\text{H}_2\text{SO}_4}$

11. A sample of oleum is labelled as $\left(100 + \frac{900}{89}\right)\%$.
Find the moles of NaOH required to completely react with 178 gm of the oleum.
(A) 4 (B) 6 (C) 3 (D) 7
12. If excess water is added into a bottle sample labelled as "112% oleum" and is reacted with 1.06 gm Na_2CO_3 , then find the volume of CO_2 evolved at 1 atm and 300 K temperature after completion of the reaction. [Assume oleum to be in excess]
(A) 12.3 L (B) 0.246 L
(C) 24.6 L (D) 2.46 L
13. The number of Nitrogen atoms, water molecules and the total charge on the product side are respectively when following reaction is balanced?
 $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2 + \text{H}_2\text{O}$
(A) 6, 3, 0 (B) 4, 2, 2 (C) 4, 2, 0 (D) 3, 2, 0
14. Which of the following statement is **CORRECT**?
(A) CHF_3 is more acidic than CHCl_3
(B) Geometry around 'N' is linear in $\text{H}_3\text{Ge-NCO}$
(C) $[\text{B}_3\text{O}_6]^{-3}$ & $\text{B}_3\text{N}_3\text{H}_6$ are planar
(D) In $\text{O}(\text{SiH}_3)_2$ bond angle only affected by steric repulsion
15. Which of the following statement is correct?
(A) Octet rule is followed by N in NO_2
(B) BF_3 is hypervalent species and PF_5 is hypovalent species
(C) Cl-O bond order in ClO_3^- is less than in ClO_4^-
(D) All of the above
16. Which of the following order is/are **CORRECT**?
(A) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3$ (Lewis acidic character)
(B) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_3^-$ (Bond angle)
(C) $:\text{CH}_2 < :\text{CF}_2 < :\text{CCl}_2$ (Stability of Carbene)
(D) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$ (C-O Bond length)
17. The nature of overlap in the bridge bond present in I_2Cl_6 is :
(A) $\text{sp}^3\text{-p-sp}^3$
(B) $\text{sp}^3\text{d}^2\text{-sp}^3\text{-sp}^3\text{d}^2$
(C) $\text{sp}^3\text{d-sp}^3\text{-sp}^3\text{d}$
(D) $\text{sp}^3\text{d-sp}^2\text{-sp}^3\text{d}$
18. According to M.O.T., which of the following molecule has maximum number of π -bond?
(A) C_2 (B) O_2
(C) OF (D) NO
19. Which of the following is not a correct order
(i) $\text{AlCl}_3 < \text{AlBr}_3 < \text{AlI}_3$ (Covalent character)
(ii) $\text{Rb}_2\text{CO}_3 > \text{K}_2\text{CO}_3 > \text{Na}_2\text{CO}_3$ (decomposition temperature)
(iii) $\text{BeC}_2\text{O}_4 < \text{MgC}_2\text{O}_4 < \text{CaC}_2\text{O}_4$ (Solubility)
(iv) $\text{Li}_3\text{N} > \text{Na}_3\text{N} > \text{K}_3\text{N}$ (Thermal stability)
(A) i, iii, iv
(B) i, iii
(C) only iii
(D) iii and iv

20. Solid AB form NaCl type structure in which A^+ ion is present at corner & face center of unit cell. If this unit cell is cut in two identical parts according to given plane in the figure. These two parts of unit cell called newly generated unit cells. INCORRECT option is -



- (A) Empirical formula of AB remains unchanged in newly generated unit cell.
- (B) Equal numbers of A^+ and B^- ions are present at both edge centres and corners in newly generated unit cell.
- (C) Cation A^+ will be present on face centres of newly generated unit cell
- (D) Newly generated unit cell will have same lattice parameters as that of tetragonal lattice system

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

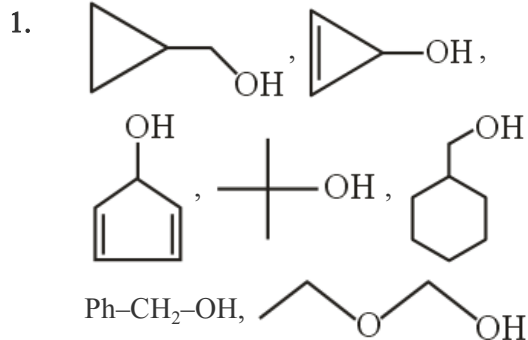
For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

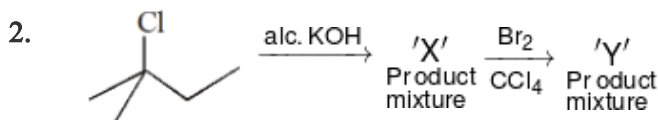
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

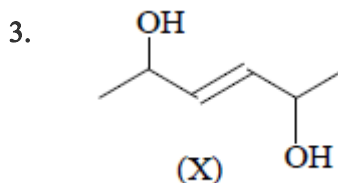
Negative Marks : -1 If wrong option is selected.



Identify total number of compound gives immediate turbidity with Lucas reagent.



Total number of possible dibromo products in final product mixture (Y)



Identify total number of stereoisomer's if double bond is present in cis configuration in given compound (X)

4. An ionic compound $\left(\overset{+}{X} \overset{-}{Y} \right)$ is doped with controlled amount of (MY_2) . Number of cation vacancies in 1 mol of XY is found to be 6.02×10^{21} . If atomic mass of X, Y and M are 15, 10 & 20 a.m.u respectively : (Assume volume remains unchanged) (Given : $N_A = 6.02 \times 10^{23}$) If the % decreases in density is x, then the value of $10x$ is :

5. 585.5 g of 0.5 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____ .
[Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$]
[Give your answer as nearest integer].
6. A mixture of H_2 , N_2 and O_2 occupying 100 ml underwent reaction so as to form H_2O_2 (l) and N_2H_2 (g) as the only products, causing the volume of contract by 60 ml. The remaining mixture was passed through pyragallol causing a contraction of 10 ml. To the remaining mixture excess H_2 was added and the above reaction was repeated, causing a reduction in volume of 10 ml. (No other products are formed)
What is the mole percent of N_2 in the initial mixture ?
7. The dipole moment of molecule A–B is 1.6D. If bond length of A–B molecule is $1.625\text{\AA}$, then %ionic character is calculated to be x%. Calculate $x \times 0.39$?
8. Total number of boron atom in anionic part of borax which form σ -coordinate bond is x and total number of boron atom which can form **$\pi\pi$ - $\pi\pi$** coordinate bond is y then calculate value of $x - y$
9. Find the number of non redox reaction to give their product(s) from the following :
(1) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$
(2) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow$
(3) $\text{N}_2\text{O}_3 + \text{H}_2\text{O} \rightarrow$
(4) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow$
(5) $\text{XeF}_4 + \text{H}_2\text{O} \rightarrow$
10. Total number of species in which highest occupied molecular orbital has gerade type symmetry.
 B_2 , N_2 , C_2 , N_2^{2+} , O_2 , O_2^{2-} , N_2^{2-}

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Number of solutions of equation $2020 \operatorname{sgn}(x^3 - 4x) + 2022 \operatorname{sgn}(x^2 - x) = 2021$, is (sgn denotes signum function)
(A) 0 (B) 2 (C) 5 (D) infinite
- Range of the function $f(x) = \frac{3 \sin x + 4 \cos x - 2}{3 \sin x + 4 \cos x + 10}$, is
(A) $\left[0, \frac{1}{5}\right]$ (B) $\left[-1, \frac{1}{5}\right]$
(C) $[-2, 1]$ (D) $\left[-\frac{7}{5}, \frac{1}{5}\right]$
- $\lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ is equal to :-
(A) 1
(B) -1
(C) Limit doesn't exist
(D) $\frac{\pi}{2}$
- Number of values of x at which $f(x) = \left|x - \frac{3}{2}\right| + |x - 2| + (x - 1)|x - 1|$ is not differentiable is :-
(A) 1 (B) 2
(C) 3 (D) 4

- If $f(x) = \begin{cases} a \cot^2 x \ln \cos x & ; x < 0 \\ 1 & ; x = 0 \\ (\sec x)^b \cot x & ; x > 0 \end{cases}$ is continuous at $x = 0$, then :-
(A) $a = -2, b \in \mathbb{R}$ (B) $a = -2, b = 0$
(C) $a = -2, b = e$ (D) $a = -2, b = \frac{1}{e}$
- Let $p, q, r \in \mathbb{R}^+$ such that $p + q + r = 21$, then the value of $\left(\frac{p^x + q^x + r^x}{3}\right)^{1/x}$ can not be equal to :-
(A) 7 (B) 4 (C) 8 (D) 5
- If $x = \cos t$ and $y = \sin t$ (where $t \neq n\pi ; n \in \mathbb{I}$), then $\frac{\left(\frac{d^2 y}{dx^2}\right)}{\left(\frac{d^2 x}{dy^2}\right)}$ is equal to :-
(A) 1 (B) $\cos^3 t$ (C) $\tan^3 t$ (D) $\cot^3 t$
- If $y = \sin x + \cos x$ and $f(x) = \frac{d^{2020} y}{dx^{2020}}$ then $y\left(\frac{\pi}{4}\right) \cdot f\left(\frac{5\pi}{4}\right)$ is :-
(A) 2 (B) -2 (C) $\sqrt{2}$ (D) 0
- If $x + y = t + \frac{1}{t}$ and $x^2 + y^2 = t^2 + \frac{1}{t^2}$ then $x^2 \frac{dy}{dx}$ is :-
(A) 1 (B) -1 (C) -2 (D) 2
- If $y = e^{\tan^{-1} x} + e^{-\cot^{-1} x}$, then $(1 + x^2)y'' + (2x - 1)y'$ is :-
(A) 0 (B) 1 (C) -1 (D) 2

11. $\frac{d}{dx} \left\{ \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\} =$
 (A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{2}$ (D) -1
12. If $f(x) = x^3 + x$ and $g(x)$ be inverse of $f(x)$, then value of $g''(-2)$ is :-
 (A) $\frac{1}{64}$ (B) 64
 (C) $\frac{32}{3}$ (D) $\frac{3}{32}$
13. If
 $f(x) = \cot^{-1}(\sqrt{x^2-1}) + \sec^{-1}\left(\frac{x}{\sqrt{x^2-1}}\right) + \cos \sec^{-1}x$,
 then $f'(-2)$ is equal to :-
 (A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$
 (C) $\frac{1}{2\sqrt{3}}$ (D) $-\frac{1}{2\sqrt{3}}$
14. Let $f(x) = x^3 + ax^2 + ax + 1$ and $\lim_{x \rightarrow \frac{1}{3}} (f(x))^{\frac{1}{3x-1}}$ has non zero finite value then sum of roots of equation $f(x) = 0$ is
 (A) $\frac{3}{4}$ (B) $-\frac{4}{3}$
 (C) $\frac{1}{12}$ (D) $-\frac{1}{12}$
15. A derivable function $f(x)$ satisfies $f(x+y) = f(x) + f(y) + 3x^2y + 3xy^2 - 2 \forall x, y \in \mathbb{R}$, $f'(0) = 1$, then $f(x)$ is :-
 (A) Many-one function
 (B) One-one function
 (C) Odd function
 (D) Even function

16. $f(x) = \lim_{n \rightarrow \infty} \frac{1 - |x| + (1 + \cos \pi x) x^{2n}}{1 + x^{2n}}$ is :-
 (A) non-differentiable at two points.
 (B) discontinuous at two points.
 (C) continuous but non-differentiable at three points.
 (D) non-differentiable at more than three points.
17. If $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$, $f_3(x) = \frac{1}{1-x}$, then find $J(x)$ such that $f_2 \circ J \circ f_1(x) = f_3(x)$?
 (A) $f_1(x)$ (B) $\frac{1}{x} f_3(x)$
 (C) $f_3(x)$ (D) $f_2(x)$
18. If r_1, r_2 , and r_3 be the radii of excircles of the triangle ABC, then $\frac{\sum r_1}{\sqrt{\sum r_1 r_2}}$ is equal to -
 (A) $\sum \cot \frac{A}{2}$ (B) $\sum \cot \frac{A}{2} \cot \frac{B}{2}$
 (C) $\sum \tan \frac{A}{2}$ (D) $\prod \tan \frac{A}{2}$
19. In a triangle ABC, if mid points of sides BC, CA & AB are D, E & F respectively, then $(AD^2 + BE^2 + CF^2) : (EF^2 + DE^2 + DF^2)$ is equal to -
 (A) 4 : 3 (B) 3 : 1
 (C) 3 : 2 (D) 8 : 3
20. The number of real solution(s) of $2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} = 3$ is-
 (A) 0 (B) 1
 (C) 3 (D) infinite

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2 \cos 2x \\ \cos 3x & \sin 3x & 3 \cos 3x \end{vmatrix}$, then $f' \left(\frac{\pi}{2} \right)$ is equal to-
2. If $f(x) = \frac{\sin x}{1+} \frac{\cos x}{1+} \frac{\sin x}{1+} \frac{\cos x}{1+} \dots \infty$, then $10f'(0)$ is
3. The function $f(x)$ satisfies the relation $f(x) \cdot f'(-x) = f(-x) \cdot f'(x) \forall x \in \mathbb{R}$ and $f(0) = 3$, then $f(3) \cdot f(-3) = ?$

4. If $\lim_{x \rightarrow 0^+} \frac{1}{(1 + \cos ecx)^{\frac{1}{\tan(\sin x)}}} = \frac{e}{k}$ then k equals to
5. Number of values of x for which $f \circ g(x) = g \circ f(x)$ where $f(x) = \frac{10^x - 1}{10^x + 1}$ & $g(x) = x + 1$, is
6. If $f(x) = \{x\} + \{2x\} + \{3x\}$ is not differentiable at 7 points in $[0, n)$, (where, $n \in \mathbb{N}$ and $\{.\}$ is fractional part function) then n is equal to :-
7. If in $\triangle ABC$ length of a side and opposite angle are 2 units & 30° respectively, then circumradius of it's pedal triangle is
8. If $2 \leq a < 3$, then the value of $\cos^{-1} \cos[a] + \cos ec^{-1} \cos ec[a] + \cot^{-1} \cot[a] = \pi + \lambda$, (where $[.]$ denotes greatest integer less than equal to x), then λ is equal to
9. If the solution of $\sin^2 x + \cot^2 x = 3$, where $x \in \left(0, \frac{\pi}{2}\right)$ is $x = k$, then $[\operatorname{cosec}^2 k]$ is equal to, where $[.]$ denotes greatest integer function
10. If $\sin^2 \theta + \sin^2 \alpha = 2 \sin \theta + 2 \sin \alpha$ where $\theta, \alpha \in [0, \pi]$, then number of possible values of $\theta + \alpha$ is equal to

Space for Rough Work

